

第三节 定积分的换元法和分部积分法

定积分的换元法

一、换元公式

定理 假设 (1) $f(x)$ 在 $[a, b]$ 上连续;

(2) 函数 $x = \varphi(t)$ 在 $[\alpha, \beta]$ 上是单值的且有连续导数;

(3) 当 t 在区间 $[\alpha, \beta]$ 上变化时, $x = \varphi(t)$ 的值在 $[a, b]$ 上变化, 且 $\varphi(\alpha) = a$ 、 $\varphi(\beta) = b$,

则 有
$$\int_a^b f(x) dx = \int_{\alpha}^{\beta} f[\varphi(t)] \varphi'(t) dt.$$

证 设 $F(x)$ 是 $f(x)$ 的一个原函数,

$$\int_a^b f(x)dx = F(b) - F(a),$$

$$\therefore \Phi(t) = F[\varphi(t)],$$

$$\Phi'(t) = \frac{dF}{dx} \cdot \frac{dx}{dt} = f(x)\varphi'(t) = f[\varphi(t)]\varphi'(t),$$

$\therefore \Phi(t)$ 是 $f[\varphi(t)]\varphi'(t)$ 的一个原函数.

$$\int_{\alpha}^{\beta} f[\varphi(t)]\varphi'(t)dt = \Phi(\beta) - \Phi(\alpha),$$

$$\varphi(\alpha) = a, \quad \varphi(\beta) = b,$$

$$\Phi(\beta) - \Phi(\alpha) = F[\varphi(\beta)] - F[\varphi(\alpha)]$$

$$= F(b) - F(a),$$

$$\int_a^b f(x) dx = F(b) - F(a) = \Phi(\beta) - \Phi(\alpha)$$

$$= \int_{\alpha}^{\beta} f[\varphi(t)] \varphi'(t) dt.$$

注意 当 $\alpha > \beta$ 时, 换元公式仍成立.

应用换元公式时应注意:

- (1) 用 $x = \varphi(t)$ 把变量 x 换成新变量 t 时, 积分限也相应的改变.
- (2) 求出 $f[\varphi(t)]\varphi'(t)$ 的一个原函数 $\Phi(t)$ 后, 不必还原成变量 x 的函数, 只要把 t 的上、下限分别代入 $\Phi(t)$ 然后相减就行了.

应用换元公式时应注意:

(3) $x = \varphi(t)$ 要求是单值函数。

(4) 定积分换元法可以用来证明积分等式，
关键在于 $x = \varphi(t)$ 的构造。这与积分的上下限
以及被积函数的形式有关。

例1 计算 $\int_0^{\frac{\pi}{2}} \cos^5 x \sin x dx$.

解 令 $t = \cos x$, $dt = -\sin x dx$,

$$x = \frac{\pi}{2} \Rightarrow t = 0, \quad x = 0 \Rightarrow t = 1,$$

$$\int_0^{\frac{\pi}{2}} \cos^5 x \sin x dx$$

$$= -\int_1^0 t^5 dt = \left. \frac{t^6}{6} \right|_0^1 = \frac{1}{6}.$$

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例2 计算 $\int_0^{\pi} \sqrt{\sin^3 x - \sin^5 x} dx$.

解 $\because f(x) = \sqrt{\sin^3 x - \sin^5 x} = |\cos x|(\sin x)^{\frac{3}{2}}$

$$\therefore \int_0^{\pi} \sqrt{\sin^3 x - \sin^5 x} dx = \int_0^{\pi} |\cos x|(\sin x)^{\frac{3}{2}} dx$$

$$= \int_0^{\frac{\pi}{2}} \cos x (\sin x)^{\frac{3}{2}} dx - \int_{\frac{\pi}{2}}^{\pi} \cos x (\sin x)^{\frac{3}{2}} dx$$

$$= \int_0^{\frac{\pi}{2}} (\sin x)^{\frac{3}{2}} d \sin x - \int_{\frac{\pi}{2}}^{\pi} (\sin x)^{\frac{3}{2}} d \sin x$$

$$= \frac{2}{5} (\sin x)^{\frac{5}{2}} \Big|_0^{\frac{\pi}{2}} - \frac{2}{5} (\sin x)^{\frac{5}{2}} \Big|_{\frac{\pi}{2}}^{\pi} = \frac{4}{5}.$$

例3 计算 $\int_{\sqrt{e}}^{e^{\frac{3}{4}}} \frac{dx}{x\sqrt{\ln x(1-\ln x)}}.$

解 原式 $= \int_{\sqrt{e}}^{e^{\frac{3}{4}}} \frac{d(\ln x)}{\sqrt{\ln x(1-\ln x)}}$

$$= \int_{\sqrt{e}}^{e^{\frac{3}{4}}} \frac{d(\ln x)}{\sqrt{\ln x} \sqrt{(1-\ln x)}} = 2 \int_{\sqrt{e}}^{e^{\frac{3}{4}}} \frac{d\sqrt{\ln x}}{\sqrt{1-(\sqrt{\ln x})^2}}$$

$$= 2 \left[\arcsin(\sqrt{\ln x}) \right]_{\sqrt{e}}^{e^{\frac{3}{4}}} = \frac{\pi}{6}.$$

例4 1) 计算 $\int_1^4 \frac{dx}{x(1+\sqrt{x})}$

解 令 $t = \sqrt{x}$, 则 $x = t^2, dx = 2t dt$,

$$\text{原式} = \int_1^2 \frac{2dt}{t(1+t)} = 2 \int_1^2 \left(\frac{1}{t} - \frac{1}{1+t} \right) dt$$

$$= 2[\ln t - \ln(1+t)]_1^2 = 2 \ln \frac{4}{3}$$

例4 2) 计算 $\int_0^1 x\sqrt{1-x} dx$

解: 令 $t = \sqrt{1-x}, x = 1-t^2, dx = -2t dt$

$$\text{原式} = \int_1^0 t(1-t^2)(-2t) dt = \int_0^1 (2t^2 - 4t^4) dt = \frac{4}{15}$$

例5 计算 $\int_0^a \frac{1}{x + \sqrt{a^2 - x^2}} dx$. ($a > 0$)

解 令 $x = a \sin t$, $0 \leq t \leq \frac{\pi}{2}$ $dx = a \cos t dt$,

$$x = a \Rightarrow t = \frac{\pi}{2}, \quad x = 0 \Rightarrow t = 0,$$

$$\text{原式} = \int_0^{\frac{\pi}{2}} \frac{a \cos t}{a \sin t + \sqrt{a^2 (1 - \sin^2 t)}} dt$$

$$= \int_0^{\frac{\pi}{2}} \frac{\cos t}{\sin t + \cos t} dt = \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(1 + \frac{\cos t - \sin t}{\sin t + \cos t} \right) dt$$

$$= \frac{1}{2} \cdot \frac{\pi}{2} + \frac{1}{2} \left[\ln |\sin t + \cos t| \right]_0^{\frac{\pi}{2}} = \frac{\pi}{4}.$$

例 6 当 $f(x)$ 在 $[-a, a]$ 上连续, 则

$$(1) \int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$$

(2) $f(x)$ 为偶函数, 则

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx;$$

(3) $f(x)$ 为奇函数, 则 $\int_{-a}^a f(x) dx = 0$.

证 $\int_{-a}^a f(x)dx = \int_{-a}^0 f(x)dx + \int_0^a f(x)dx,$

在 $\int_{-a}^0 f(x)dx$ 中令 $x = -t,$

$$\int_{-a}^0 f(x)dx = -\int_a^0 f(-t)dt = \int_0^a f(-t)dt,$$

$$\int_{-a}^0 f(x)dx = -\int_a^0 f(-t)dt = \int_0^a f(-t)dt,$$

① $f(x)$ 为偶函数, 则 $f(-t) = f(t)$,

$$\begin{aligned}\int_{-a}^a f(x)dx &= \int_{-a}^0 f(x)dx + \int_0^a f(x)dx \\ &= 2\int_0^a f(t)dt;\end{aligned}$$

② $f(x)$ 为奇函数, 则 $f(-t) = -f(t)$,

$$\int_{-a}^a f(x)dx = \int_{-a}^0 f(x)dx + \int_0^a f(x)dx = 0.$$

例7 计算 $\int_{-1}^1 \frac{2x^2 + x \cos x}{1 + \sqrt{1 - x^2}} dx.$

解 原式 = $\int_{-1}^1 \underbrace{\frac{2x^2}{1 + \sqrt{1 - x^2}}}_{\text{偶函数}} dx + \int_{-1}^1 \underbrace{\frac{x \cos x}{1 + \sqrt{1 - x^2}}}_{\text{奇函数}} dx$

$$= 4 \int_0^1 \frac{x^2}{1 + \sqrt{1 - x^2}} dx = 4 \int_0^1 \frac{x^2(1 - \sqrt{1 - x^2})}{1 - (1 - x^2)} dx$$

$$= 4 \int_0^1 (1 - \sqrt{1 - x^2}) dx = 4 - \underbrace{4 \int_0^1 \sqrt{1 - x^2} dx}_{\text{单位圆的面积}}$$

$$= 4 - \pi.$$

例 8 若 $f(x)$ 在 $[0,1]$ 上连续, 证明

$$(1) \int_0^{\frac{\pi}{2}} f(\sin x) dx = \int_0^{\frac{\pi}{2}} f(\cos x) dx; \text{ 提示: } \sin\left(\frac{\pi}{2} - t\right) = \cos t$$

$$\text{由此计算 } \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx \left(= \frac{\pi}{4} \right)$$

$$(2) \int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx.$$

$$\text{由此计算 } \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx.$$

证 (1) 设 $x = \frac{\pi}{2} - t \Rightarrow dx = -dt,$

$$x = 0 \Rightarrow t = \frac{\pi}{2}, \quad x = \frac{\pi}{2} \Rightarrow t = 0,$$

$$\int_0^{\frac{\pi}{2}} f(\sin x) dx = -\int_{\frac{\pi}{2}}^0 f\left[\sin\left(\frac{\pi}{2} - t\right)\right] dt$$

$$= \int_0^{\frac{\pi}{2}} f(\cos t) dt = \int_0^{\frac{\pi}{2}} f(\cos x) dx;$$

(2) 设 $x = \pi - t \Rightarrow dx = -dt,$

$$x = 0 \Rightarrow t = \pi, \quad x = \pi \Rightarrow t = 0,$$

$$\int_0^{\pi} x f(\sin x) dx = -\int_{\pi}^0 (\pi - t) f[\sin(\pi - t)] dt$$

$$= \int_0^{\pi} (\pi - t) f(\sin t) dt,$$

$$\begin{aligned}
 \int_0^{\pi} x f(\sin x) dx &= \pi \int_0^{\pi} f(\sin t) dt - \int_0^{\pi} t f(\sin t) dt \\
 &= \pi \int_0^{\pi} f(\sin x) dx - \int_0^{\pi} x f(\sin x) dx, \\
 \therefore \int_0^{\pi} x f(\sin x) dx &= \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx.
 \end{aligned}$$

$$\begin{aligned}
 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx &= \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx \\
 &= -\frac{\pi}{2} \int_0^{\pi} \frac{1}{1 + \cos^2 x} d(\cos x) = -\frac{\pi}{2} [\arctan(\cos x)]_0^{\pi} \\
 &= -\frac{\pi}{2} \left(-\frac{\pi}{4} - \frac{\pi}{4} \right) = \frac{\pi^2}{4}.
 \end{aligned}$$

例9 若 $f(x)$ 是以 T 为周期的连续函数, 对任意的 a 有

$$\int_a^{a+T} f(x)dx = \int_0^T f(x)dx$$

由此得,

$$\int_0^T f(x)dx = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x)dx;$$

$$\int_0^{nT} f(x)dx = n \int_0^T f(x)dx.$$

二、小结

定积分的换元法

$$\int_a^b f(x)dx = \int_{\alpha}^{\beta} f[\varphi(t)]\varphi'(t)dt$$

几个特殊积分、定积分的几个等式

备用习题:

1、设 $f(x)$ 是连续函数, 且 $\int_0^{x^3-1} f(t)dt = x$,

则 $f(7) = ?$.

$$\text{解} \left(\int_0^{x^3-1} f(t)dt \right)' = (x)' \Rightarrow f(x^3-1) \cdot 3x^2 = 1 \Rightarrow f(x^3-1) = \frac{1}{3x^2}$$

$$\text{令: } x^3 - 1 = 7 \quad \text{解得 } x=2, \text{ 所以 } f(7) = \frac{1}{3 \times 2^2} = \frac{1}{12}$$

2、设 $f(x)$ 是连续函数，且 $f(x) = x + 2\int_0^1 f(t)dt$ ，
求 $f(x)$

解：设 $\int_0^1 f(t)dt = I$ ，于是 $f(x) = x + 2I$ ，

两边在 $[0, 1]$ 上积分

$$\int_0^1 f(x)dx = \int_0^1 xdx + 2I \int_0^1 dx,$$

$$\text{即：} I = \frac{1}{2} + 2I, I = -\frac{1}{2} \quad \therefore f(x) = x - 1.$$

3. 设 $f(x) = x^2 - x \int_0^2 f(x) dx + 2 \int_0^1 f(x) dx$, 求 $f(x)$.

解 定积分为常数, 故应用积分法定此常数.

设 $\int_0^1 f(x) dx = a$, $\int_0^2 f(x) dx = b$, 则

$$f(x) = x^2 - bx + 2a$$

$$a = \int_0^1 f(x) dx = \left[\frac{x^3}{3} - \frac{bx^2}{2} + 2ax \right]_0^1 = \frac{1}{3} - \frac{b}{2} + 2a$$

$$b = \int_0^2 f(x) dx = \left[\frac{x^3}{3} - \frac{bx^2}{2} + 2ax \right]_0^2 = \frac{8}{3} - 2b + 4a$$

$$\Rightarrow a = \frac{1}{3}, b = \frac{4}{3} \Rightarrow f(x) = x^2 - \frac{4}{3}x + \frac{2}{3}$$

4、计算定积分 $\int_{-2}^2 (|x| + x)e^{-|x|} dx$

解： $|x|e^{-|x|}$ 为偶函数 $xe^{-|x|}$ 为奇函数

$$\text{原式} = 2 \int_0^2 xe^{-x} dx$$

$$= (-2xe^{-x} - 2e^{-x}) \Big|_0^2$$

$$= 2 - \frac{6}{e^2}$$

5、设 $f(x) \in C[a,b]$, $f(x)$ 在 $[a,b]$ 内可导, 且 $f(a) = 0$,
 $f(b) = \int_b^a f(t)dt$, 求证 $\exists \xi \in (a,b)$ 使 $f'(\xi) = -f(\xi)$.

解 构造辅助函数

$$F(x) = f(x) + \int_a^x f(t)dt$$

6、 设 $f(x) = \begin{cases} 1-x^2, & x \leq 0, \\ e^{-x}, & x > 0, \end{cases}$ 求 $\int_1^3 f(x-2)dx$

解 令 $x-2=t$,

$$\text{原式} = \int_{-1}^1 f(t)dt = \int_{-1}^0 (1-t^2)dt + \int_0^1 e^{-t}dt = \frac{7}{3} - \frac{1}{e}$$

7、 设 $f(x)$ 为连续函数, 求

$$(1) \quad \frac{d}{dx} \int_0^{x^2} (x^2 - t)f(t)dt \qquad 2x \int_0^{x^2} f(t)dt$$

$$(2) \quad \frac{d}{dx} \int_1^2 f(x+t)dt \qquad f(2+x) - f(1+x)$$

8. 设 $f(x)$ 可导, 且 $f(0) = 0$, $F(x) = \int_0^x t^{n-1} f(x^n - t^n) dt$

$$\text{求证: } \lim_{x \rightarrow 0} \frac{F(x)}{x^{2n}} = \frac{1}{2n} f'(0)$$

证明 令 $u = x^n - t^n$, $du = -nt^{n-1} dt$, $t : \Big|_0^x \rightarrow u : \Big|_{x^n}^0$

$$F(x) = \int_0^x t^{n-1} f(x^n - t^n) dt = \frac{1}{n} \int_0^{x^n} f(u) du$$

$$F'(x) = x^{n-1} f(x^n)$$

$$\therefore \lim_{x \rightarrow 0} \frac{F(x)}{x^{2n}} = \lim_{x \rightarrow 0} \frac{F'(x)}{2nx^{2n-1}} = \frac{1}{2n} \lim_{x \rightarrow 0} \frac{f(x^n)}{x^n}$$

$$= \frac{1}{2n} \lim_{x \rightarrow 0} \frac{f(x^n) - f(0)}{x^n - 0} = \frac{1}{2n} f'(0)$$

9、设 $f(x)$ 为连续函数， $I = t \int_0^{\frac{s}{t}} f(xt) dx$,

求 $\frac{dI}{dx}$ $\frac{dI}{dt}$ $\frac{dI}{ds}$

$$\frac{dI}{dx} = 0 \quad \frac{dI}{dt} = 0 \quad \frac{dI}{ds} = f(s)$$

关键在于换元：

x	0	$\frac{s}{t}$
u	0	s

$$xt = u, \quad t \int_0^{\frac{s}{t}} f(xt) dx = t \int_0^s f(u) \frac{1}{t} du = \int_0^s f(u) du$$

10、设 $f(x)$ 为连续函数, $F(t) = \int_1^t \left(\int_y^t f(x) dx \right) dy$, 求 $F'(2)$.

解: 由 $f(x)$ 为连续函数知 $f(x)$ 的原函数存在, 设 $G'(x) = f(x)$. 则 $\int_y^t f(x) dx = G(t) - G(y)$,
从而 $F(t) = \int_1^t \left(\int_y^t f(x) dx \right) dy = \int_1^t (G(t) - G(y)) dy = \int_1^t G(t) dy - \int_1^t G(y) dy$
 $= G(t)(t-1) - \int_1^t G(y) dy$,
于是 $F'(t) = G'(t)(t-1) + G(t) - \left(\int_1^t G(y) dy \right)' = G'(t)(t-1) + G(t) - G(t) = G'(t)(t-1)$,
故 $F'(2) = G'(2) = f(2)$.

9. 设 $f(x)$ 在 $[0, a]$ 上连续, 在 $(0, a)$ 内可导, 且 $3 \int_{\frac{2a}{3}}^a f(x) dx = f(0)a$. 证明: 存在 $\xi \in (0, a)$ 使 $f'(\xi) = 0$.

10. 设 $f(x)$ 在 $[0, 2]$ 上连续, 在 $(0, 2)$ 内二阶可导, 且 $2 \int_{\frac{1}{2}}^1 f(x) dx =$

$f(2), f\left(\frac{1}{2}\right) = f(0)$. 证明: 存在 $\xi \in (0, 2)$ 使 $f''(\xi) = 0$.

证明：用积分中值定理和罗尔定理

44 求可导函数 $f(x)$, 使它满足 $\int_0^1 f(tx) dt = f(x) + x \sin x$.

解: 令 $u = tx$, 则 $du = x dt$. 从而 $\int_0^1 f(tx) dt = \int_0^x f(u) \cdot \frac{1}{x} du = \frac{1}{x} \int_0^x f(u) du$,
于是 $\int_0^1 f(tx) dt = f(x) + x \sin x$ 可化为 $\frac{1}{x} \int_0^x f(u) du = f(x) + x \sin x$, 即为
 $\int_0^x f(u) du = x f(x) + x^2 \sin x$, 对该等式两边关于 x 求导可得 $f(x) = f(x) + x f'(x) + 2x \sin x$
 $+ x^2 \cos x$, 即有 $f'(x) = -2 \sin x - x \cos x$, 从而 $f(x) = \int (-2 \sin x - x \cos x) dx = -2 \int \sin x dx$
 $- \int x \cos x dx = \cos x - x \sin x + C$.

3. 设 $f(x) \in C[a, b]$, 且 $f(x)$ 单调上升,

$$\text{证明: } \int_a^b xf(x)dx \geq \frac{a+b}{2} \int_a^b f(x)dx.$$

证法1 构造变上限积分

$$\text{Let } F(x) = \int_a^x xf(x)dx - \frac{a+x}{2} \int_a^x f(x)dx$$

$$\text{Then } F(a) = 0$$

$$\begin{aligned} F'(x) &= xf(x) - \frac{1}{2} \int_a^x f(x)dx - \frac{a+x}{2} f(x) \\ &= \frac{x-a}{2} f(x) - \frac{1}{2} \int_a^x f(x)dx \geq \frac{x-a}{2} f(x) - \frac{x-a}{2} f(x) = 0 \end{aligned}$$

$$\therefore F(x) \geq 0 \quad x \in [a, b], \quad F(b) \geq 0 \quad \text{holds}$$

3. 设 $f(x) \in C[a, b]$, 且 $f(x)$ 单调上升,

$$\text{证明: } \int_a^b xf(x)dx \geq \frac{a+b}{2} \int_a^b f(x)dx.$$

证法2 $\int_a^b (x - \frac{a+b}{2}) f(x) dx =$

$$= \int_a^{\frac{a+b}{2}} (x - \frac{a+b}{2}) f(x) dx + \int_{\frac{a+b}{2}}^b (x - \frac{a+b}{2}) f(x) dx$$

$$= f(\xi_1) \int_a^{\frac{a+b}{2}} (x - \frac{a+b}{2}) dx + f(\xi_2) \int_{\frac{a+b}{2}}^b (x - \frac{a+b}{2}) dx$$

$$= \frac{(a+b)^2}{2} [f(\xi_2) - f(\xi_1)] \geq 0$$

$$\xi_1 \in [a, \frac{a+b}{2}], \xi_2 \in [\frac{a+b}{2}, b]$$

定积分的分部积分法

一、分部积分公式

设函数 $u(x)$ 、 $v(x)$ 在区间 $[a, b]$ 上具有连续导数,

$$\text{则 } \int_a^b uv' dx = [uv]_a^b - \int_a^b vu' dx, \text{ 或 } \int_a^b u dv = [uv]_a^b - \int_a^b v du.$$

定积分的分部积分公式

推导 $(uv)' = u'v + uv', \quad \int_a^b (uv)' dx = [uv]_a^b,$

$$[uv]_a^b = \int_a^b u'v dx + \int_a^b uv' dx, \quad \therefore \int_a^b uv' dx = [uv]_a^b - \int_a^b u'v dx,$$

$$\text{或 } \int_a^b u dv = [uv]_a^b - \int_a^b v du.$$

例1 计算 $\int_0^{\frac{1}{2}} \arcsin x dx$.

解 令 $u = \arcsin x$, $v' dx = dx$,

$$\text{则 } du = \frac{dx}{\sqrt{1-x^2}}, \quad v = x,$$

$$\int_0^{\frac{1}{2}} \arcsin x dx = [x \arcsin x]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \frac{x dx}{\sqrt{1-x^2}}$$

$$= \frac{1}{2} \cdot \frac{\pi}{6} + \frac{1}{2} \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}} d(1-x^2)$$

$$= \frac{\pi}{12} + \left[\sqrt{1-x^2} \right]_0^{\frac{1}{2}} = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1.$$

例2 计算 $\int_0^{\frac{\pi}{4}} \frac{x dx}{1 + \cos 2x}$.

解 $\because 1 + \cos 2x = 2 \cos^2 x,$

$$\therefore \int_0^{\frac{\pi}{4}} \frac{x dx}{1 + \cos 2x} = \int_0^{\frac{\pi}{4}} \frac{x dx}{2 \cos^2 x} = \int_0^{\frac{\pi}{4}} \frac{x}{2} d(\tan x)$$

$$= \frac{1}{2} [x \tan x]_0^{\frac{\pi}{4}} - \frac{1}{2} \int_0^{\frac{\pi}{4}} \tan x dx$$

$$= \frac{\pi}{8} - \frac{1}{2} [\ln \sec x]_0^{\frac{\pi}{4}} = \frac{\pi}{8} - \frac{\ln 2}{4}.$$

例3 计算 $\int_0^1 \frac{\ln(1+x)}{(2+x)^2} dx$.

解 $\int_0^1 \frac{\ln(1+x)}{(2+x)^2} dx = -\int_0^1 \ln(1+x) d\frac{1}{2+x}$

$$= -\left[\frac{\ln(1+x)}{2+x} \right]_0^1 + \int_0^1 \frac{1}{2+x} d\ln(1+x)$$

$$= -\frac{\ln 2}{3} + \int_0^1 \boxed{\frac{1}{2+x} \cdot \frac{1}{1+x}} dx \rightarrow \frac{1}{1+x} - \frac{1}{2+x}$$

$$= -\frac{\ln 2}{3} + [\ln(1+x) - \ln(2+x)]_0^1 = \frac{5}{3}\ln 2 - \ln 3.$$

注意的问题

初等函数的原函数不一定是初等函数,因此不一定都能积出.

例如, $\int e^{-x^2} dx$, $\int \frac{\sin x}{x} dx$, $\int \sin x^2 dx$,
 $\int \frac{1}{\ln x} dx$, $\int \frac{dx}{\sqrt{1+x^4}}$, $\int \sqrt{1+x^3} dx$,
 $\int \sqrt{1-k^2 \sin^2 x} dx$ ($0 < k < 1$), $\dots\dots$

例4 设 $f(x) = \int_1^{x^2} \frac{\sin t}{t} dt$, 求 $\int_0^1 xf(x)dx$.

解 因为 $\frac{\sin t}{t}$ 没有初等形式的原函数,
无法直接求出 $f(x)$, 所以采用分部积分法

$$\int_0^1 xf(x)dx = \frac{1}{2} \int_0^1 f(x) d(x^2)$$

$$= \frac{1}{2} [x^2 f(x)]_0^1 - \frac{1}{2} \int_0^1 x^2 df(x)$$

$$= \frac{1}{2} f(1) - \frac{1}{2} \int_0^1 x^2 f'(x) dx$$

$$\because f(x) = \int_1^{x^2} \frac{\sin t}{t} dt, \quad f(1) = \int_1^1 \frac{\sin t}{t} dt = 0,$$

$$f'(x) = \frac{\sin x^2}{x^2} \cdot 2x = \frac{2 \sin x^2}{x},$$

$$\therefore \int_0^1 x f(x) dx = \frac{1}{2} f(1) - \frac{1}{2} \int_0^1 x^2 f'(x) dx$$

$$= -\frac{1}{2} \int_0^1 2x \sin x^2 dx = -\frac{1}{2} \int_0^1 \sin x^2 dx^2$$

$$= \frac{1}{2} [\cos x^2]_0^1 = \frac{1}{2} (\cos 1 - 1).$$

例5 证明定积分公式

$$I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx$$

$$= \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}, & n \text{ 为正偶数} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{4}{5} \cdot \frac{2}{3}, & n \text{ 为大于1的正奇数} \end{cases}$$

证 设 $u = \sin^{n-1} x$, $dv = \sin x dx$,

$$du = (n-1) \sin^{n-2} x \cos x dx, \quad v = -\cos x,$$

$$I_n = \left[\underbrace{-\sin^{n-1} x \cos x}_0 \right]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \underbrace{\cos^2 x}_{1 - \sin^2 x} dx$$

$$I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$= (n-1)I_{n-2} - (n-1)I_n$$

$$I_n = \frac{n-1}{n} I_{n-2} \quad \text{积分 } I_n \text{ 关于下标的递推公式}$$

$$I_{n-2} = \frac{n-3}{n-2} I_{n-4} \quad \dots\dots, \text{直到下标减到0或1为止}$$

$$I_{2m} = \frac{2m-1}{2m} \cdot \frac{2m-3}{2m-2} \cdots \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} I_0, \quad (m = 1, 2, \cdots)$$

$$I_{2m+1} = \frac{2m}{2m+1} \cdot \frac{2m-2}{2m-1} \cdots \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} I_1,$$

$$I_0 = \int_0^{\frac{\pi}{2}} dx = \frac{\pi}{2}, \quad I_1 = \int_0^{\frac{\pi}{2}} \sin x dx = 1,$$

于是
$$I_{2m} = \frac{2m-1}{2m} \cdot \frac{2m-3}{2m-2} \cdots \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2},$$

$$I_{2m+1} = \frac{2m}{2m+1} \cdot \frac{2m-2}{2m-1} \cdots \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3}.$$

思考题

设 $f''(x)$ 在 $[0,1]$ 上连续, 且 $f(0)=1$,
 $f(2)=3$, $f'(2)=5$, 求 $\int_0^1 xf''(2x)dx$.

思考题解答

$$\begin{aligned}\int_0^1 x f''(2x) dx &= \frac{1}{2} \int_0^1 x df'(2x) \\&= \frac{1}{2} [x f'(2x)]_0^1 - \frac{1}{2} \int_0^1 f'(2x) dx \\&= \frac{1}{2} f'(2) - \frac{1}{4} [f(2x)]_0^1 \\&= \frac{5}{2} - \frac{1}{4} [f(2) - f(0)] = 2.\end{aligned}$$

二、小结

定积分的分部积分公式

$$\int_a^b u dv = [uv]_a^b - \int_a^b v du.$$

(注意与不定积分分部积分法的区别)